Practical Class Week 1

Aims:

We will be using a variety of maps in thus unit and this class aims to introduce you to working with maps. We will use map scales and grid references to locate features on a map of the campus.

What are the intended learning outcomes? At the end of this class you should be able to:

1. Estimate the distances between features on a map using the correct map scale
2. Record the grid reference for named features on a map
3. Plot the positions of features on a field map

Equipment:

Mapping board, paper map of Brunswick Park, notebook, pencil, pen, phone or tablet with OS locate app, time chart (if you have an iphone ICS Timechart app)

Activities:

You should have read the information on the BB page about maps and watched the short videos.

We will have a short introduction to the activities before we go outside.

When we go outside you can work in groups of 2-5 people, or you can work on your own.

You do not have to do the activities in order – please start where there is space. There are lots of people here to help you, there are 4 lecturers and 4 Graduate Teaching Assistants, so if you need help or do not understand anything ask us – we are here to help you. We will be wearing high viz jackets so you can find us.

Scale:

1. Look at your paper copy of the map of Brunswick Park
    a. What is the scale of this map?

    b. What would 10 metres on the ground measure on the map?

    c. What distance does a line of 2cm on the map represent on the ground?

    d. Work out how long is your stride length

Using your strides to measure the distance work out

    e. What is the distance between the doors of the Williamson Building and the Simon Building

f. What is the distance between the Simon Building and the Zochonis Building – mark on the map where you have measured this.

g. What is the total length of Brunswick Park –how did you estimate this?

Grid References (GR):

The university campus is found within the SJ 100km grid square of the UK National Grid

1. Find where the Grid reference shown below is on the map. This is a 12 figure GR – it will locate you within a 1m square.
   Easting **= 384775** Northing = **396699**
   Walk to this GR – what is the feature you can see on this building? How might this help increase biodiversity in Brunswick Park?

   a. What does the OS Locate app say is the GR for this locality? How accurate is this GR?

2. Because we are using a large scale map we do not need to give the full GR, we only need to use the GR for within each 100m square. So, walk to **GR 468667**. Remember the Easting is given first in the GR and then the Northing. This example is a 6 figure GR – Easting = 468 Northing = 667

   How many trees can you see? How might you show their distribution and other characteristics?

3. What is the GR for the North East corner of the Williamson Building?

4. What is the GR for the entrance to the Museum?

5. What is the GR for South West corner of the Kilburn Building?

Mapping the timeline (revision from Tuesday)

Use your estimated distance for the length of Brunswick Park and use this to represent the period of time from the formation of the Earth (4.56 billion years)

Length of Brunswick Park =

What amount of time is represented by 1m?

What distance will represent 10 million years?

Draw a line on your map to represent the timeline (it should be the length you estimated for the length of Brunswick Park)

On your timeline mark where the beginning of the Archean, the Carboniferous, the Jurassic and the Paleogene occur.

What are the 6 and 8 figure grid references for the 6 points on your timeline?

Origin of the Earth =

Archean =

Carboniferous =

Jurassic =

Paleogene =

Present day =